EXPERIMENT 2-CLI, GUI AND VUI

◇ CLI (Command Line Interface)

CLI allows users to interact with the system using text commands typed in the terminal. It is fast, lightweight, and suitable for experienced users.

CODE:

```
import os import

sys


def rename_file(old_name, new_name):

try:

        os.rename(old_name,  new_name)              print(f"File

renamed  from  {old_name}  to  {new_name}")         except

FileNotFoundError:

        print(f"Error: {old_name} not found.")

except Exception as e:       print(f"An error

occurred: {e}")


if __name__ == "__main__":

if len(sys.argv) != 3:

        print("Usage: python rename_file_cli.py <old_filename>
<new_filename>")

else:
```

rename_file(sys.argv[1], sys.argv[2])

OUTPUT:

```
Usage: python rename_file_cli.py <old_filename> <new_filename>
PS C:\Users\ADMIN\OneDrive\Desktop\UID> python rename_file_cli.py
 final.txt old.txt
File renamed from final.txt to old.txt
```

HOW IT WORKS:

- User selects create / rename / exit from menu
- Program performs file operation using Python functions
- Output displayed in terminal

◇ GUI (Graphical User Interface)

GUI allows users to interact using graphical elements like buttons, text boxes, and windows. It is user-friendly and easy to use.

CODE:

import tkinter as tk from tkinter

import messagebox import os

def rename_file():

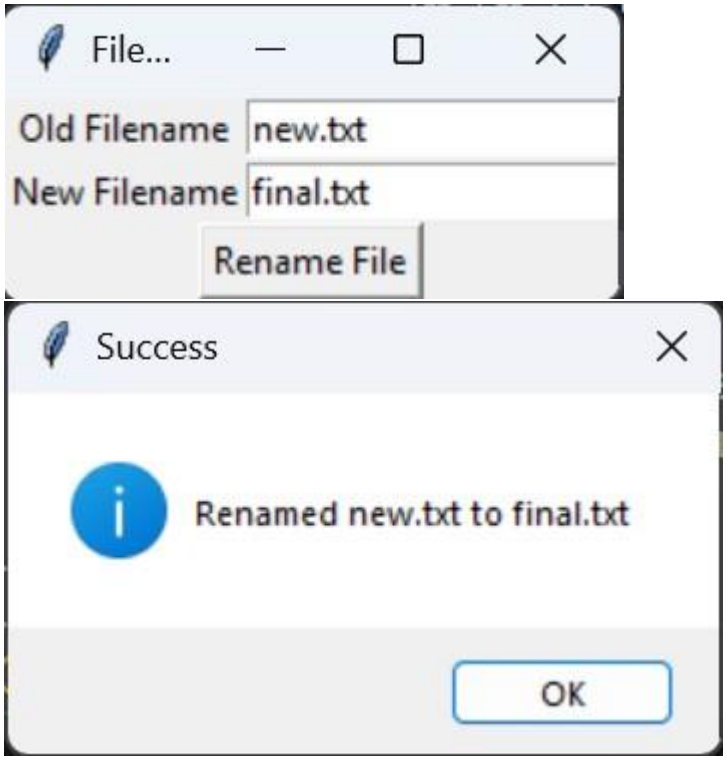
old_name = old_entry.get()

new_name = new_entry.get() try:

os.rename(old_name, new_name)
messagebox.showinfo("Success", f"Renamed {old_name} to

```
{new_name}")     except
FileNotFoundError:
        messagebox.showerror("Error", "File not found")

root = tk.Tk() root.title("File
Renamer")

tk.Label(root, text="Old Filename").grid(row=0, column=0) tk.Label(root,
text="New Filename").grid(row=1, column=0)

old_entry = tk.Entry(root) new_entry
= tk.Entry(root)
old_entry.grid(row=0, column=1)
new_entry.grid(row=1, column=1)

tk.Button(root, text="Rename File", command=rename_file).grid(row=2,
columnspan=2)

root.mainloop()
```

OUTPUT:



HOW IT WORKS:

- User enters file name in text box
- Click buttons (Create, Rename, Delete)
- Files shown in listbox
- Status bar shows result

◇ VUI (Voice User Interface)

VUI allows users to interact with the system using voice commands. It improves accessibility and provides hands-free operation.

CODE:

```
import speech_recognition as sr import
os
import sounddevice as sd from
scipy.io.wavfile import write

samplerate = 44100

def record(seconds=4):
print("Speak now...")
    audio = sd.rec(int(seconds * samplerate), samplerate=samplerate,
channels=1, dtype='int16')     sd.wait()     write("temp.wav",
samplerate, audio)

def listen(prompt):
print(prompt)     record()

    r = sr.Recognizer()     with
sr.AudioFile("temp.wav") as source:
        audio_data = r.record(source)

try:
        text = r.recognize_google(audio_data)
print("You said:", text)         return
text.lower()     except:
        print("Could not understand. Try again.")
return None
```

```
def rename_file():          print("Welcome to the Voice-
Controlled File Renamer!")     old_name = listen("Say the
name of the file you want to rename
(without .txt)")     if
not old_name:
        return

    new_name = listen("Say the new name for the file (without .txt)")
if not new_name:        return

    old_file = old_name + ".txt"
new_file = new_name + ".txt"

    if not os.path.exists(old_file):
        print(f"Error: '{old_file}' not found.")
return

    os.rename(old_file, new_file)

    # GREEN SUCCESS LINE WITH TICK
    print(f"\033[92m☑ File successfully renamed from '{old_file}' to
'{new_file}'\033[0m")

rename_file()
```

OUTPUT:

```
Welcome to the Voice-Controlled File Renamer!
Say the name of the file you want to rename (without .txt)
Speak now...
You said: final
Say the new name for the file (without .txt)
Speak now...
You said: voice
✅ File successfully renamed from 'final.txt' to 'voice.txt'
```

HOW IT WORKS:

- System listens to voice command
- User says create or rename
- Program performs operation
- System speaks result